

Nomenclature

C_1, C_2	coefficients in one-dimensional radial conduction equation
q''	heat flux
r	radial coordinate
t	time
T	temperature
T_f	bulk liquid temperature
T_s	heater surface temperature
ΔT_{sat}	excess heater surface temperature, $T_s - T_{\text{sat}}$
ΔT_{sub}	liquid subcooling, $T_{\text{sat}} - T_f$

Greek symbol

θ	orientation angle
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Subscripts

f	bulk liquid
sat	saturation

reservoir is vented to the ambient to prevent any pressure fluctuations in the system due to the vapor production.

Fig. 2(b) shows detailed dimensions of an individual heater. To construct the heaters, a plate of oxygen-free copper is turned on a lathe to form a ring with the diameters and thicknesses shown. The ring is then cut into multiple pieces such that the bulk of each copper piece is a section spanning 8.8° , except for the bottom (wetted) surface, which spans 10.0° . When the nine heaters are assembled together, as illustrated in Fig. 2(a), the heater construction minimizes circumferential heat flow between heaters while producing a continuous boiling surface. For reference purposes, the heaters are numbered sequentially from 1 (bottom-most heater) to 9 (top heater). Heat losses are minimized by sandwiching the heaters between G-10 insulation, and covering the top portions of the heaters with fiberglass batting, as illustrated in Fig. 2(c). Heat is supplied from a thick film resistor that is silver-soldered onto each copper heater. The nine heaters are powered by a single 120 V a.c. variac.

heater locations along the vertical axis, and the locations of the teleconverter in conjunction with the heater locations.

Uncertainty in the heater locations is determined by the current transducer reading, the data presented in this paper, and the measurements. The uncertainty in the heat flux and the aforementioned temperature is $\pm 2.6^\circ\text{C}$. Heat loss through the walls of the measured electrical circuit is negligible.

3. Experimental results

3.1. Nucleate boiling and critical heat flux

Fig. 3 shows boiling curves for the two heaters. Heater 1 is a degree downward-facing, and heater 2 is a degree upward-facing. Portion of the convex surface of the heaters 1 and 2, inciting nucleate boiling, is $\Delta T_{\text{sat}} < 2^\circ\text{C}$, rather than $\Delta T_{\text{sat}} < 1^\circ\text{C}$ for the flat surfaces in Fluorinerts [5]. The heat fluxes for heaters 1 and 2 are then proportional to the surface area of the heaters. The boiling regime is similar for both heaters, with the surface temperature associated with the first, followed by the other heaters.

It is clear from Fig. 3 that the boiling regime is similar, while heater 1 is the hottest. As the temperature increased, the degradation of the heater surface becomes more pronounced. The degradation is more pronounced on these heaters due to the lack of additional cooling available from the buoyancy-driven liquid-vapor interface.

Stable heat fluxes appear for both heaters, denoted as CHF-, and heater 1 is the hottest. Fig. 3(b) shows the temperature excursion. Notice that the temperature rise is rapid once initiated. However, once initiated, the temperature rise is as large as at CHF-. At approximately 100 s in the temperature-time plot, these two heaters are essentially at the same temperature. The "leak" into heater 3, causing the temperature to rise, is negligible.